

Point Calibration Does Not Control Type I Error Across the Composite Null in a Composed Adaptive Bayesian Design

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JSM 2026 • Session: AI, Machine Learning & Digital Tools in Clinical Development • August 3, 2026

Abstract

Efficacy thresholds for Bayesian monitoring can be calibrated at a single assumed control rate. We examined whether such point calibration controls Type I error (T1E) across the composite null in a two-arm Phase II design with historical control borrowing, Bayesian monitoring, sample size re-estimation (SSR), and response-adaptive randomization (RAR). Each component passed its numerical benchmark check, but none of those checks evaluated T1E away from the calibration anchor.

The full-design threshold was calibrated to $\alpha = 0.025$ at $p_0 = 0.25$ with no time trend. We then evaluated a one-dimensional null slice spanning common baseline rates $p = 0.20$ – 0.36 , with a shared trend active in both arms. SSR used the blinded pooled response rate only as a nuisance-parameter estimate while holding the planned treatment effect fixed. With all mechanisms active, T1E rose from 0.0112 to 0.2272. At the same fixed threshold, disabling SSR and RAR did not restore control; the ablation reached 0.1638 at $p = 0.36$. The fixed-threshold difference between the full design and the ablation reversed sign across the range. The SSR $\times$ RAR interaction was $+0.0106$ (95% simulation interval $[-0.0074, +0.0286]$), and a fresh-seed estimate was $+0.0003$ ($[-0.0227, +0.0234]$); neither supports a stable super-additive interaction claim.

The grid demonstrates a control failure but does not locate the supremum. Operating characteristics should therefore be evaluated across prespecified null-generating conditions, not only at the calibration anchor.

1. Introduction

Computational tools are used to specify priors, simulate operating characteristics and prepare regulatory submissions. In the workflow examined here, the checks applied to such a tool are component-level and numerical: each adaptive mechanism is compared against a reference implementation in isolation. This paper asks what those checks do and do not establish about the error rate of the assembled design. We make no claim about how widespread that workflow is; the question is what follows from it where it is used.

The FDA’s January 2026 draft guidance on the use of Bayesian methodology in clinical trials [U.S. Food and Drug Administration, 2026] is directly relevant. The guidance recommends that sponsors document operating characteristics under realistic simulation scenarios, pre-specify how prior-data conflict will be handled, and evaluate whether Bayesian adaptive designs meet frequentist operating characteristic targets. (It is a draft guidance containing nonbinding recommendations, and we describe it as such throughout.) These recommendations bear on a specific practice, and one this paper examines directly: an efficacy threshold calibrated at a single assumed control rate, with that single point then treated as evidence of error control.

This paper makes two claims, at two levels. The primary claim is about **robustness**: an efficacy threshold calibrated at one assumed control rate does not deliver Type I error control across the composite null. The secondary claim is about **composition**: at the frozen full-design threshold, the combined SSR/RAR contrast changes sign across the evaluated null rates, but the factorial interaction estimates are too imprecise to support a stable super-additive interaction claim.

Both claims are demonstrated by simulation over a grid of common baseline rates. We are explicit about what such a grid can and cannot establish: it exhibits points at which control fails, but it does not locate or estimate the supremum of the Type I error over the null.

We present Zetyra as the reproducible evaluation infrastructure for this work. The platform is evidence and infrastructure for the argument; it is not the argument itself.

2. Background

2.1 What Component-Level Checks Establish

“Validation” of clinical trial design software is not a term with a single agreed meaning. Where it amounts to documenting that design outputs were computed with a named package, with cross-checks against published values, it does not establish maximum numerical deviation across the range of design parameters, does not guarantee reproducibility under stochastic methods, and does not account for the behaviour of multiple adaptive components operating jointly. The FDA January 2026 draft guidance raises the standard explicitly: operating characteristics should be evaluated by comprehensive simulation over plausible assumptions—the guidance discusses prior-data conflict explicitly and calls for scenarios spanning plausible values of design inputs, including the background rate of the endpoint—and documented reproducibly.

The simulations and component benchmarks are implemented in Zetyra. The versioned source, package information and seeds that produced every reported number are released with the paper (see Acknowledgements); the public validation repository contains the broader software checks.

2.2 Composed Adaptive Designs and Marginal Calibration

Definition. A *composed adaptive design* is a design in which two or more adaptive mechanisms read the same interim data before the final analysis. In the design studied here, Bayesian monitoring, the SSR trigger and the RAR allocation rule all compute from the accumulated responses at interim looks.

We do not formalise this as a set-theoretic condition on information sets. Writing $\mathcal{F}_i \cap \mathcal{F}_j \neq \emptyset$ would not distinguish anything: for sigma-fields the intersection always contains $\emptyset$ and Ω , so the condition holds trivially for any pair. Nor is the converse available—informational independence does not by itself guarantee that error control survives adaptation.

The relevant observation is weaker and purely negative. Let $\phi(X)$ denote the final rejection rule (Bayesian monitoring), and let $A_1(X_t)$, $A_2(X_t)$ denote the adaptation rules—SSR and RAR—which alter sample size or allocation based on interim data X_t but do not themselves reject H_0 . Each is validated independently (the blinded nuisance-parameter SSR satisfies its pooled-input, algebra and bound checks; RAR tracks $P(\theta_e > \theta_c)$ within tolerance). The composed decision rule is $\phi_{\text{comp}}(X) = \phi(X; A_1(X_t), A_2(X_t))$: the final rejection indicator evaluated on data whose sample size and allocation were shaped by the adaptation rules. The standard approach calibrates that composed rule at a single point, choosing γ^* so that $P_{p_0, \delta=0}(\phi_{\text{comp}}(X) = 1) \approx \alpha$ at one anchor rate

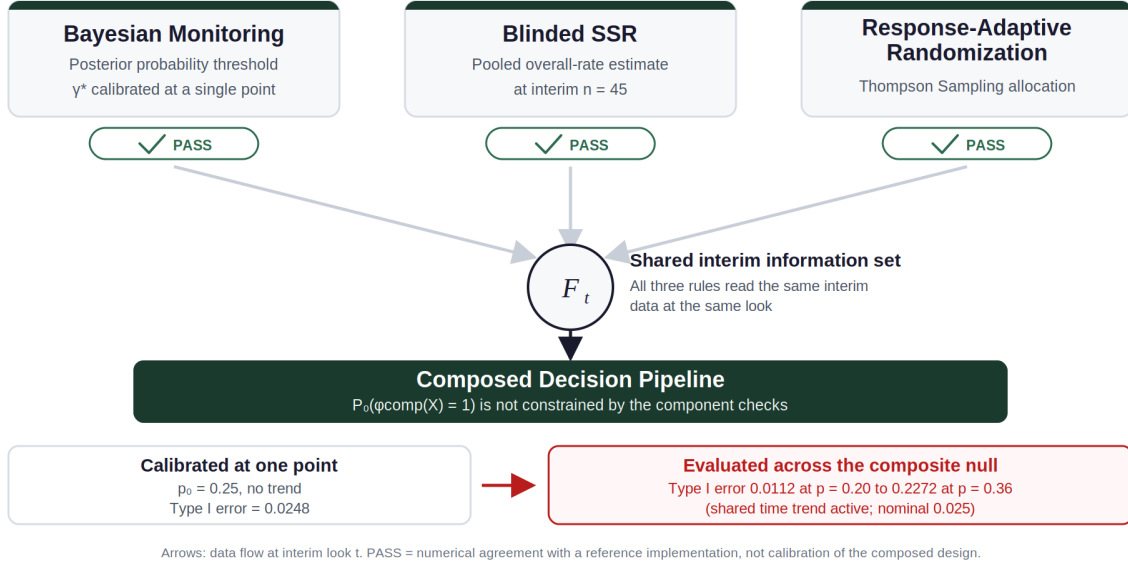


Figure 1: Composed adaptive design: Bayesian monitoring, blinded nuisance-parameter SSR and RAR share interim data before the final analysis. Component checks and point calibration at the anchor do not bound Type I error elsewhere in the composite null.

with no trend. That establishes the error rate at that point and nowhere else: it places no bound on $P_{p,\delta}(\phi_{\text{comp}}(X) = 1)$ for other (p, δ) in the null. The component benchmark checks add no such bound either, since none of them evaluates an error rate.

Calibrating each component under its own marginal model, and the composed rule at a single null point, places no constraint on the error rate of the composed rule elsewhere in the null. This is a statement about what those checks fail to establish, not a condition under which inflation must occur. Whether it occurs, in which direction and by how much are empirical questions, which is why the remainder of the paper is a simulation study. Figure 1 illustrates the structure.

3. Simulation Study

3.1 Design Specification

We simulate a two-arm Phase II oncology design with MAP borrowing on the control arm [Weber et al., 2021], interim looks at $n = 30, 45, 60$ and a final at $n \geq 90$ (post-SSR). RBesT fits the historical cohorts ($n = 50, 40, 60$; 38/150 responses) as a robustified mixture with components 0.5887 Beta(18.403, 53.273), 0.2113 Beta(1.956, 4.703) and 0.2000 Beta(1, 1). At 50% enrollment, Rule B uses only the pooled response count to estimate the overall event-rate nuisance parameter, while the planned treatment effect $\Delta = 0.45 - 0.25 = 0.20$ remains fixed. Under a prespecified working 1:1 allocation, it preserves the fixed-sample Wald power implied by the original $N = 90$ design (0.5296), permits increases only, and applies the protocol cap $N = 150$. The working rates are $\hat{p}_c = \text{clamp}(\hat{p}_{\text{pool}} - \Delta/2)$ and $\hat{p}_e = \hat{p}_c + \Delta$; the recalculated per-arm size is

$$n_{\text{arm}} = \frac{(z_{1-\alpha} + z_{\text{power}})^2 [\hat{p}_c(1 - \hat{p}_c) + \hat{p}_e(1 - \hat{p}_e)]}{\Delta^2},$$

rounded up and doubled. This pooled-data construction follows the blinded binary internal-pilot principle of Friede and Kieser [2004]. It is a conventional-style working nuisance rule, not an exact power calculation for the Thompson-sampling allocation or Bayesian final decision. At the planned $n = 45$ look, its complete count mapping spans only $N = 90$ – 100 ; the exact expected N is 90.44 at $p = 0.25$, 91.57 at $p = 0.30$, and 94.12 at $p = 0.36$ under the active trend. Thompson Sampling RAR is benchmarked against analytical $P(\theta_e > \theta_c)$.

The threshold is calibrated at the design anchor $p_0 = 0.25$, distinct from the dominant-component mean 0.2568, robustified-mixture mean 0.3132 and historical pooled rate 0.2533. We therefore describe $p > 0.25$ only as departure from the calibration anchor, not from the prior mean.

Under H_0 , both arms share baseline rate p and drift together, so the reported rejection probabilities are frequentist Type I error, not posterior predictive error. We evaluate $p = 0.27$ (mild) and $p = 0.30$ (strong) with linear trend $\delta = 0.05$ over the planned enrollment horizon. These are sensitivity scenarios, not an asserted empirical range; Tang et al. [2010] and Viele et al. [2014] provide clinical and historical-borrowing context.

All estimates carry binomial MCSE $\sqrt{\hat{p}(1 - \hat{p})/N_{\text{sim}}}$; cell-specific MCSEs and 95% simulation intervals are reported here or in the simulation supplement.

Repeated nominal conditions use independent seeds and simulation budgets; their largest difference is about 2.0 combined standard errors. We report each in its originating experiment rather than pooling it.

Note on calibration: The efficacy threshold is obtained by grid-free binary search, giving $\gamma^* = 0.96566$ with in-sample T1E = 0.0248 (MCSE 0.0011, $N = 20,000$) at the design anchor $p_0 = 0.25$ with no trend. On an independent fresh-seed sample ($N = 5,000$), T1E was 0.0224 (MCSE 0.0021; 95% simulation interval [0.0183, 0.0265]), compatible with the nominal target. Binary search still targets an empirical sample rather than the population error rate, so neither estimate is exact calibration.

3.2 Component-Level Benchmark Checks

Each adaptive component is individually benchmarked. MAP prior: maximum deviation from RBesT mixture = 1.78×10^{-15} (floating-point precision). Rule B SSR: the calculation reproduces $N = 90$ at the exact planning nuisance rate, uses only the pooled interim count, is increase-only, returns even sample sizes, and stays within the prespecified bounds; its full $n = 45$ input support was exhaustively checked. The pooled nuisance construction is compared with the framework of Friede and Kieser [2004], but the embedded working Wald rule is not claimed to reproduce the operating characteristics of their final test. RAR one-step allocation: maximum deviation from analytical $P(\theta_e > \theta_c) = 0.0036$ (Monte Carlo error at $n = 100,000$). All three components pass their benchmark checks; none of those checks evaluates Type I error away from the anchor.

3.3 Type I Error Across the Composite Null

We evaluate a one-dimensional slice of the composite null, fixing $\delta = 0.05$ and varying the common baseline rate p . Both arms share the same response process, so H_0 holds throughout. No row is the calibration scenario: even the $p = 0.25$ row has an active trend, whereas γ^* was tuned at $\delta = 0$.

p	SSR + RAR disabled	All mechanisms	Fixed- γ^* difference
0.20	0.0336	0.0112	-0.0224
0.22	0.0428	0.0232	-0.0196
0.25	0.0738	0.0490	-0.0248
0.27	0.0810	0.0698	-0.0112
0.30	0.1174	0.1166	-0.0008
0.33	0.1404	0.1810	+0.0406
0.36	0.1638	0.2272	+0.0634

Two readings follow. First, **at the full-design threshold, disabling SSR and RAR does not restore control**: with both switched off, T1E still rises from 0.0336 at $p = 0.20$ to 0.1638 at $p = 0.36$. That column is the same fixed-threshold ablation used throughout (Section 3.4)—it carries γ^* , which was calibrated with SSR and RAR active. At $p = 0.25$ on this trend-active slice its T1E is 0.0738; neither that row nor the full-design row is the no-trend calibration scenario. Read it as “the design with the adaptive components switched off, threshold unchanged”, not as a calibrated monitoring-only design. Second, **at fixed γ^* , enabling SSR and RAR lowers T1E at low p and raises it at high p** : the difference runs from -0.0224 at $p = 0.20$ to $+0.0634$ at $p = 0.36$, reversing sign across the range.

The grid establishes a direction change, not its crossing point: the contrast at $p = 0.30$ is within Monte Carlo error and the next evaluated rate is 0.33. It also demonstrates failure of uniform control but does not locate the supremum; it establishes only that the supremum is at least 0.2272, the largest observed value. The more aggressive Rule A stress test is summarized in Appendix A.

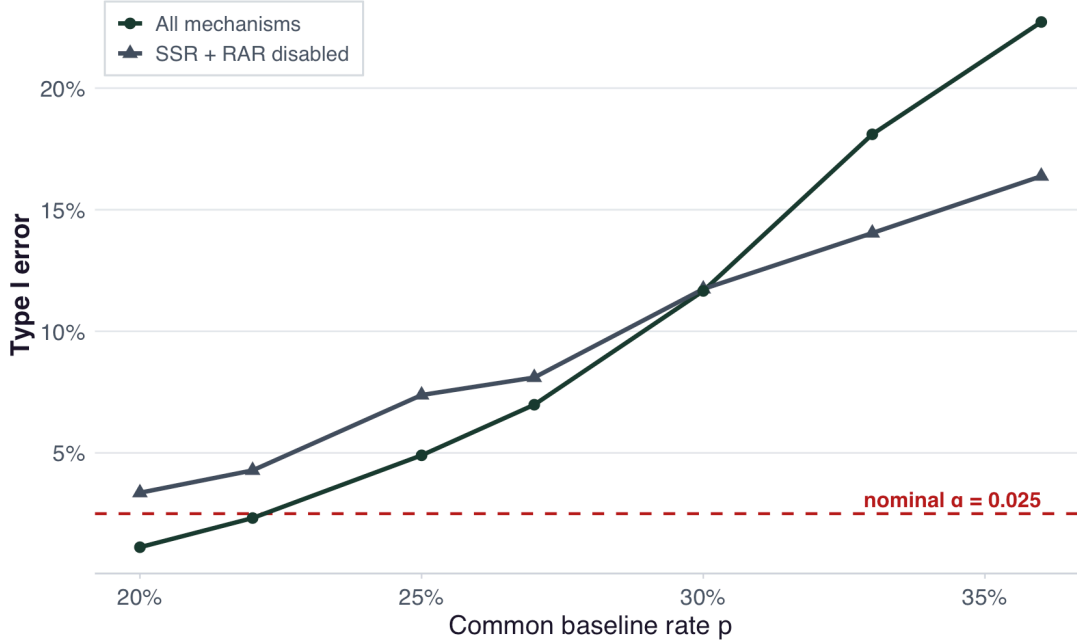


Figure 2: Type I error along the composite-null slice $\delta = 0.05$. Grey holds the full-design threshold with SSR and RAR disabled; dark green shows all mechanisms. Dashed red line: nominal $\alpha = 0.025$.

3.4 Fixed-Threshold Factorial Ablation: SSR by RAR

Four designs (SSR $\times$ RAR), all at $p = 0.30$ with the trend active:

Condition	T1E	MCSE
SSR + RAR disabled	0.1170	0.0045
+ SSR only	0.1138	0.0045
+ RAR only	0.1202	0.0046
+ SSR and RAR	0.1276	0.0047

These are ablations at a fixed threshold, not a decomposition. γ^* was calibrated with SSR and RAR active; disabling components while retaining it does not yield separately calibrated designs or attributable shares. The baseline cell is therefore not “the excess caused by monitoring”, although it does show that disabling both components fails to restore control.

The simple contrasts are $\Delta_{\text{SSR}|\text{RAR}=0} = -0.0032$ and $\Delta_{\text{RAR}|\text{SSR}=0} = +0.0032$; the interaction is $\Delta_{\text{interaction}} = +0.0106$ (propagated MCSE 0.0092, 95% SI $[-0.0074, +0.0286]$). A fresh-seed repeat ($N = 3,000$ per cell) gave $+0.0003$ (MCSE 0.0118; 95% SI $[-0.0227, +0.0234]$). Both intervals include zero. The point estimates therefore do not support a stable SSR-by-RAR interaction claim under Rule B. Joint evaluation remains necessary because an interaction is possible, but its sign and magnitude should not be inferred from these runs.

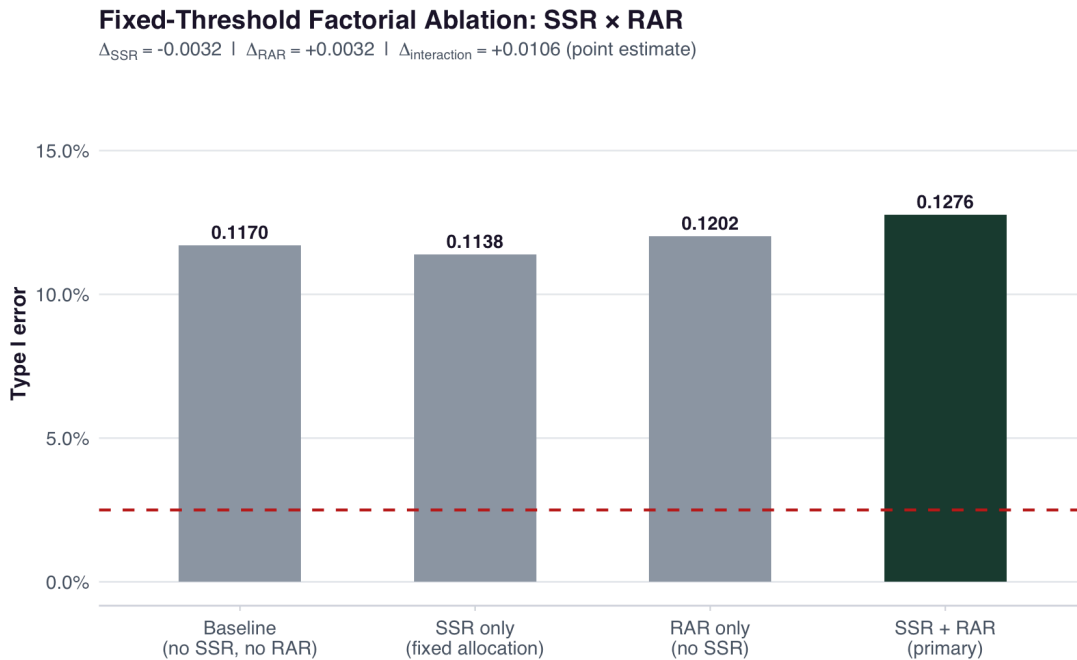


Figure 3: Fixed-threshold Rule B factorial ablation at $p = 0.30$ with the shared trend active. The interaction annotation is a point estimate; its propagated 95% Monte Carlo simulation interval includes zero, as does the fresh-seed repeat. Dashed red line = nominal $\alpha = 0.025$.

3.5 Baseline-Rate Departure by Time Trend

A 2×2 factor crossing evaluates departure and trend under the primary design; both factors feed monitoring, SSR and RAR, so it does not isolate those channels.

Condition	T1E	vs. calibration
Neither (calibration)	0.0236	—
Departure only	0.0894	+0.0658
Trend only	0.0468	+0.0232
Both (primary)	0.1206	+0.0970
Additive prediction	0.1126	
Interaction term	+0.0080	

The departure-only simple contrast is +0.0658, and the trend-only simple contrast is +0.0232. Because both factors feed monitoring, SSR and RAR, neither contrast is attributable to a single channel. Unlike the fixed-threshold ablations, all four cells share the full design and a common calibrated baseline. The interaction is +0.0080 (MCSE 0.0071; 95% SI $[-0.0060, +0.0220]$), so it does not demonstrate super-additivity. It is a risk-difference interaction at these factor levels and includes the additional trend exposure from enrollment beyond $N = 90$; Rule B limits that extension to 100 at the planned SSR look.

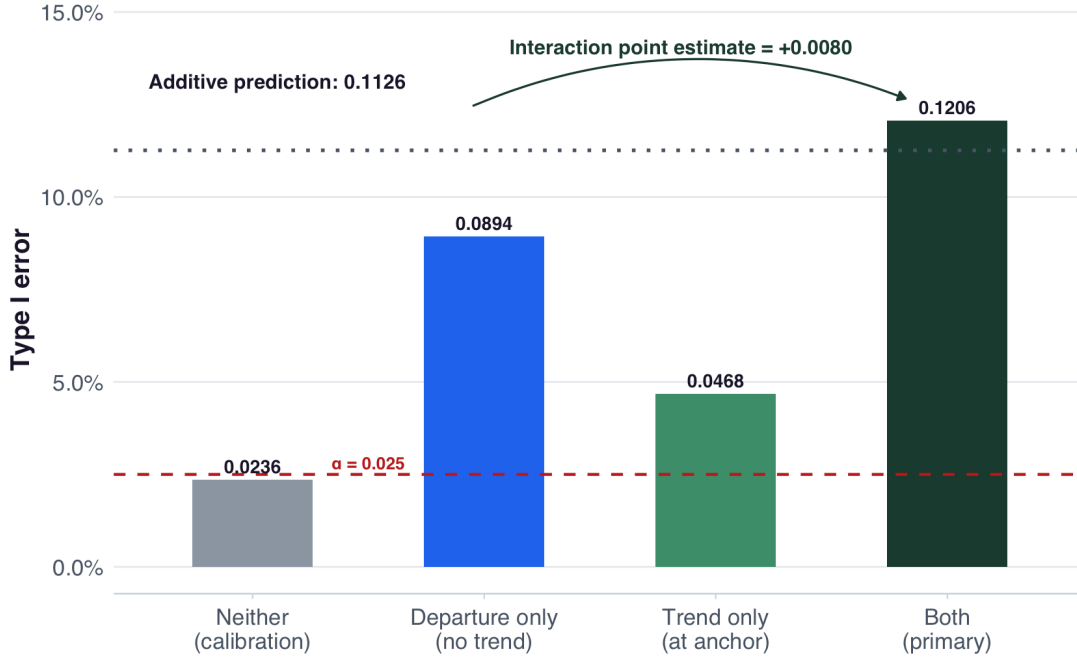


Figure 4: Departure-by-trend crossing under the full design. The interaction is +0.0080 (95% SI $[-0.0060, +0.0220]$); dashed red line denotes nominal $\alpha = 0.025$.

3.6 Sensitivity Analyses

The following sweeps use the strong departure parameterisation ($p = 0.30$) to characterize mechanism sensitivity across a wider range.

- **Departure severity**, at constant trend ($p \in \{0.25, 0.27, 0.30\}$, all with $\delta = 0.05$): T1E increases monotonically, $0.0490 \rightarrow 0.0698 \rightarrow 0.1166$, with no direction reversal.
- **Vague-component weight** (0–90%): larger weight on the flat robustify component means less borrowing; T1E declines from 0.1488 to 0.0624.
- **Time trend δ** (0–15%): T1E increases from 0.0812 to 0.2220.
- **SSR trigger timing**: moving the primary look from $n = 45$ to $n = 30$ changed T1E from 0.1161 to 0.1244; a separate sweep over $n_{\text{SSR}} \in \{20, 30, 40, 45, 55, 60\}$ ranged from 0.1126 to 0.1234. Timing also changes the realised final- N distribution, so these are not isolated timing effects.

4. Discussion

4.1 Regulatory Implications and Recommendations

An efficacy threshold calibrated at one assumed control rate need not control Type I error elsewhere in the composite null. Consistent with the FDA January 2026 draft guidance, operating-characteristic evaluation should therefore span realistic null-generating conditions and pre-specify how prior–data conflict will be handled.

For this design class, sponsors should (1) evaluate a grid of plausible control rates rather than one anchor; (2) label whether component ablations are separately calibrated, since a fixed-threshold ablation tests restoration of control rather than attribution; and (3) cross departure and time trend rather than rely only on separate one-factor analyses.

4.2 Mitigation Strategies

Potential safeguards include more robust or dynamic borrowing, pre-specified prior–data disagreement diagnostics, pipeline-level recalibration of γ^* over the null grid, and an explicit maximum acceptable T1E over a stated robustness range. Each changes the composed pipeline and must itself be evaluated at pipeline level; no component-level argument guarantees restored control.

4.3 Limitations and Future Work

This paper demonstrates a control failure by simulation in a specific design class; several limitations bound what it establishes.

The supremum is not estimated. The grid exhibits points at which control fails and establishes that the supremum is at least 0.2272, the largest value observed. It does not locate the worst case.

Conflict is not quantified. We vary the calibration-anchor departure, not a prior-predictive conflict measure. Extending prior–data diagnostics such as Matthews and Qian [2026], Qian and Matthews [2026] from Normal-mean settings to this binary MAP-mixture design would require a new statistic and calibration.

Rule B is a working nuisance calculation, not exact design power. Its 1:1 Wald model differs from the RAR allocation and Bayesian final decision; the 0.5296 reference power and 90–100 mapping follow from the original $N = 90$ design. Results are conditional on this prespecified rule. Appendix A summarizes the more aggressive Rule A stress test.

No sufficient condition is proved. Shared information neither guarantees nor precludes error control; a conditional-error or design-specific argument would be required.

Calibration is Monte Carlo. Binary search targets the empirical calibration sample; a fresh-seed validation was compatible with nominal, but neither finite simulation establishes the population error rate exactly.

Component ablations are not separately calibrated. They retain the full-design γ^* and answer whether disabling a component restores control, not how much excess it caused. Attribution would require calibrating each design separately.

5. Conclusion

Point calibration did not provide uniform Type I error control across the composite null. At the full-design threshold, disabling SSR and RAR did not restore control. The fixed- γ^* contrast reversed sign across the evaluated null rates, but the Rule B factorial and departure-by-trend interaction intervals both included zero; the simulations do not support a stable super-additive interaction claim.

The practical consequence is a reporting standard rather than a new method. Operating characteristics should be evaluated over a prespecified grid of null-generating conditions. Fixed-threshold ablations can show whether disabling a component restores control, but attribution requires separately calibrating each design. The simulation code is publicly available for independent verification and extension; the Zetyra platform is a commercial SaaS product.

Calibrate at a point. Validate over the null.

Acknowledgements

Simulation source, manuscript source, figures and reproducibility instructions:
github.com/evidenceinthewild/Composed-Adaptive-Trial-Validation. Rule A is summarized in Appendix A as a method-development stress test, not as the primary analysis.

A. Original Rule A Pooled Conditional-Power Stress Test

The original Rule A treated the pooled interim rate as a one-sample signal against $p_0 = 0.25$ and increased total sample size to target 80% conditional power, capped at $N = 150$. A common upward nuisance shift under the two-arm null could therefore drive expansion, so it was replaced before publication by Rule B. With $\gamma^* = 0.96596$, Rule A reached T1E 0.2618 at $p = 0.36$ under the trend-active null grid, compared with 0.2272 under Rule B. This more aggressive result is retained only as a stress-test summary; full outputs and source are not part of the primary Rule B release.

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